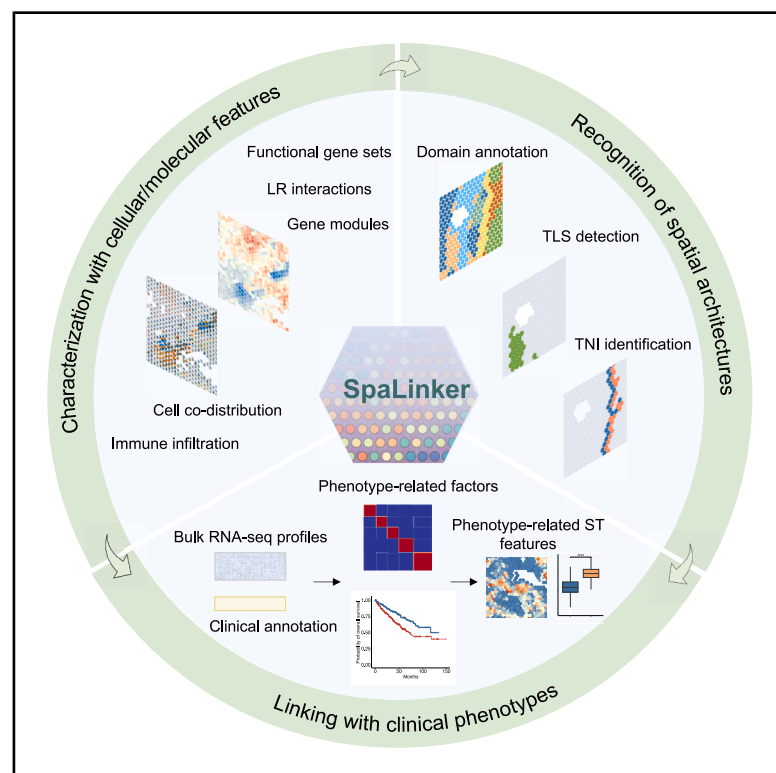


SpaLinker identifies phenotype-associated spatial tumor microenvironment features by integrating bulk and spatial sequencing data

Graphical abstract



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In brief

Xiaojie Cheng et al. introduce SpaLinker, a novel framework that integrates spatial transcriptomics with bulk omics annotations to link tumor microenvironment features with clinical outcomes. This approach enhances understanding of tumor complexity, revealing clinically relevant structures and spatial features crucial for advancing cancer diagnostics and treatment strategies.

Highlights

- SpaLinker characterizes molecular and cellular features of the TME
- SpaLinker identifies spatial architectures within the TME
- SpaLinker links spatially resolved TME features to clinical phenotypes
- Phenotype-associated TME features are identified *de novo* across multiple cancers



Technology

SpaLinker identifies phenotype-associated spatial tumor microenvironment features by integrating bulk and spatial sequencing data

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<https://doi.org/10.1016/j.xgen.2025.100893>

SUMMARY

The emergence of spatial transcriptomics (ST) technology offers unprecedented opportunities to elucidate the complexity and heterogeneity of the tumor microenvironment (TME). However, quantitatively linking spatially resolved features with clinical phenotypes remains challenging due to the scarcity of clinical annotations of spatial sequencing samples. Herein, we introduce SpaLinker, an innovative integrated framework that utilizes ST data to decipher spatially resolved TMEs at molecular, cellular, and tissue structure levels. Specifically, it assesses the prognostic significance of spatially defined features by integrating well-accumulated bulk RNA sequencing (RNA-seq) data, using a phenotype-driven computational framework. Applying SpaLinker to diverse tumor ST datasets demonstrated its utility and effectiveness in recognizing spatial architectures, including tertiary lymphoid structures and tumor-normal interfaces, and in establishing links to distinct clinical outcomes. Overall, this study presents a valuable and comprehensive pan-cancer analytical platform to *de novo* identify phenotype-associated spatial TME features, significantly enhancing the clinical utility of spatial sequencing technology.

INTRODUCTION

Tumor tissue is a highly organized and complex ecosystem consisting of malignant tumor cells and various components of the tumor microenvironment (TME), which shape various tumor structures and are closely linked to tumor development and patient prognosis.^{1,2} The recently emerged spatial transcriptomics (ST) technology^{3–7} provides an unprecedented way to explore the TME and characterize spatial heterogeneity. Integrating ST data with a single-cell RNA sequencing (scRNA-seq) reference enables the exploration of the TME at high resolution.^{8–14}

Various spatial TME features can be explored using ST technologies. First, molecular expression and cellular interactions play a crucial role in TME remodeling, leading to distinct functional or physical structures. Second, tumor architectures or compartments can be detected by various domain segmentation methods^{15–19}; however, identifying specialized structures like tertiary lymphoid structure (TLS) and tumor-normal interface

(TNI) regions remains a challenge. Although several upstream analysis tools, such as Giotto,²⁰ SPATA,²¹ and Squidpy,²² facilitate the processing and basic analysis of various ST data, there is still a lack of an effective framework that can quantitatively characterize the TME and effectively identify spatial TME features using ST.

More importantly, spatial features derived from expression and coordinate data are associated with treatment response and prognosis.²³ For example, multiple studies have shown that tumor cells and normal cells located at the leading edges display elevated levels of unique molecules that are linked to immunotherapy response or patient prognosis.^{24–27} Unfortunately, given the scarcity of available clinical annotations for spatial omics, the challenge of further interpreting the intricate relationship between spatial characteristics of the TME and patient clinical outcomes remains unaddressed. Traditional methods typically focus on regional marker genes in ST to interpret clinical associations. However, they rely on pre-defined region annotations,



limiting their ability to *de novo* identify phenotype-related markers in an unbiased and comprehensive manner. Recently, computational tools^{28,29} have been developed to identify phenotype-specific cell subpopulations from single-cell data. Nevertheless, these tools fail to capture spatial information and do not effectively illustrate the complex mechanisms that coexist within samples.

To this end, we present SpaLinker, an innovative and integrated framework that utilizes ST data to decipher spatial TME features at the molecular, cellular, and tissue structure levels and link them with clinical phenotypes, including the following: (1) SpaLinker quantitatively deciphers the molecular and cellular features of the TME; (2) SpaLinker effectively identifies spatial architectures of tumor tissues, including TLS and TNI regions; and (3) importantly, SpaLinker *de novo* identifies phenotype-related spatial features by integrating the rich clinical annotation information from well-accumulated bulk RNA-seq data using a phenotype-driven strategy. Through investigation of distinct tumor ST datasets, we demonstrate the general utility and effectiveness of SpaLinker in characterizing the TME in a pan-cancer perspective, particularly in the *de novo* linking of spatially resolved TME features with clinical phenotypes by integrating bulk and spatial sequencing data. Further applications in renal cell carcinoma (RCC), hepatocellular carcinoma (HCC), and melanoma demonstrate that SpaLinker is capable of *de novo* revealing clinical discoveries without any spatial omics clinical annotations.

DESIGN

Overview of SpaLinker

SpaLinker is an integrative analytical framework designed for ST data, aiming to identify and link spatially resolved TME features with clinical phenotypes by integrating bulk and spatial sequencing data. It consists of three functional parts (Figure 1): (1) characterization of the TME with molecular and cellular features, including assessing the enrichment of annotated functional gene sets, determining spatial gene coexpression modules, quantifying the ligand-receptor (LR) interactions, calculating co-distribution scores of cell pairs, and measuring immune infiltration levels; (2) recognition of the TME architectures, including spatial domain and TLS and TNI regions; and (3) *de novo* linking spatial TME features with clinical phenotypes without spatial omics clinical annotations. Specifically, to address the challenge of limited clinical annotation of ST data, SpaLinker designs a non-negative matrix factorization (NMF)-based technology to identify phenotype-related factors by leveraging rich clinical annotation information from well-accumulated bulk RNA-seq data and further links TME characteristics with patient clinical phenotypes by recovering these factors within ST data.

RESULTS

SpaLinker identifies the molecular and cellular levels of TME features

The TME is characterized by a highly complex and dynamic interplay that occurs among a vast array of molecules and diverse populations of cells. SpaLinker identifies the molecular

and cellular levels of the TME features, including the following: (1) it assesses the enrichment or activity levels of annotated functional gene sets. SpaLinker enables intuitive characterization of the TME by facilitating exploration of well-defined gene annotation sets, including publicly available TME-related signatures and various biomedical ontologies sourced from databases (Figure 2A; Table S1). (2) It determines gene coexpression modules. SpaLinker employs weighted gene coexpression network analysis (WGCNA)³⁰ to identify genes with similar expression patterns in space, which often share the same biological functions (Figure 2B). (3) It quantifies LR interactions. SpaLinker enables interaction establishment between LR pairs from the CellPhoneDB database³¹ by considering the effect interaction distance of secreted proteins and membrane proteins separately (Figures 2C and 2D; STAR Methods). (4) It describes the spatial distribution patterns of cell populations by calculating codistribution scores of cell pairs and measures the degree of immune infiltration using two measures: immune enrichment and immune diversity (Figure 2E; STAR Methods).

SpaLinker identifies the spatial architecture levels of TME features

SpaLinker accurately identifies TLS across cancers

SpaLinker utilizes gene expression, cell localization, and spatial information from ST data, which precisely meet the requirements for TLS characterization, to detect the presence of TLSs. Specifically, SpaLinker designs a feature selection procedure to determine predictive features (Figures 3A and 3B; STAR Methods), together with a stepwise identification procedure for TLS recognition. As a result, the gene set LC.50sig,³² which demonstrated superior predictive power among the seven TLS signatures (Figure 3B), and the codistribution of plasma/B cells and T cells were integrated with spatial information to improve TLS identification (Figure 3C; STAR Methods). The superiority of SpaLinker in TLS identification was validated using a well-annotated RCC dataset.³³ With the ground truth of validated TLS, we found that the unit-integrated features consistently outperformed their original values in all the samples (Figures 3D, 3E, and S1). We also compared the distribution difference and performance between adjusted and unadjusted values. Notably, the adjusted unit-integrated values exhibited a clearer data distribution of TLS scores without compromising performance (Figures 3D and S2; Table S2). Furthermore, it is evident that the predicted TLS scores achieved better predictive capability compared to the mean of single features, and the TLS regions identified by SpaLinker showed a high degree of consistency with the ground truth of RCC samples (Figures 3F, 3G, S3A–S3D, and S4A). Additionally, we examined the TLS scores using six TLS-negative samples, where only limited or minimal TLS signals were detected (Figure S3E).

In addition, we conducted independent validation using additional cancer types, including HCC³² ($n = 4$), intrahepatic cholangiocarcinoma³² (ICC; $n = 1$), combined hepatocellular and cholangiocarcinoma³² (cHC; $n = 1$), breast cancer³⁴ (BRCA; $n = 1$), and nasopharyngeal carcinoma³⁵ (NPC; $n = 4$). In the cases of HCC, ICC, and cHC, the TLSs were confirmed and labeled by pathologists in the original study. We found that the TLSs identified by SpaLinker achieved high accuracy,

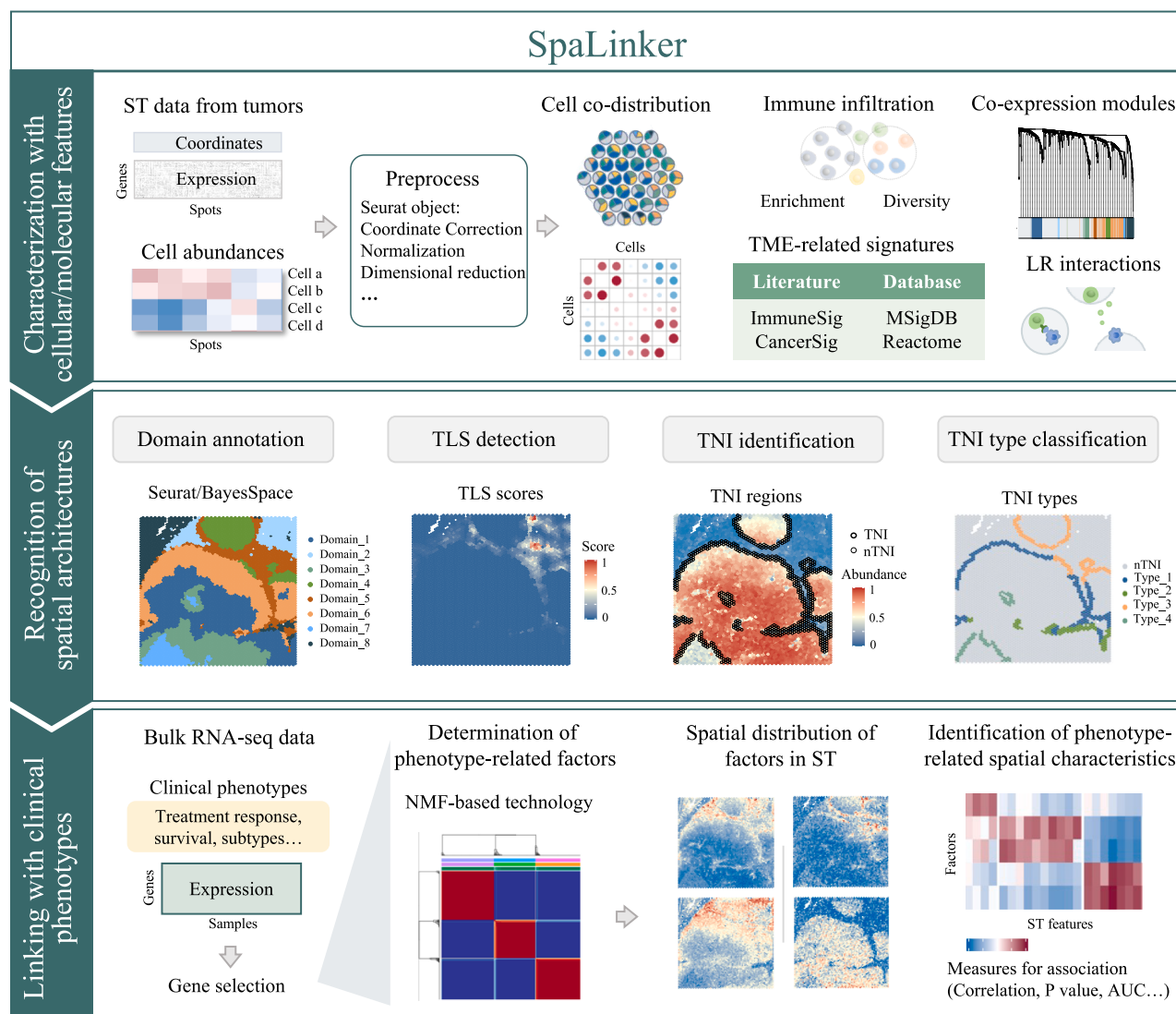


Figure 1. Framework of SpaLinker

The framework of SpaLinker consists of three parts: (1) characterization of the TME with cellular and molecular features, (2) recognition of tumor spatial structure features, and (3) *de novo* linking spatially resolved TME features with clinical phenotypes by integrating bulk and spatial sequencing data.

with a precision-recall area under the curve (PR-AUC) from 0.76 to 0.86 and a precision for TLS spots from 0.56 to 0.85 (Figures 3H and S4B). In the BRCA and NPC samples, the TLS areas marked in images were annotated by pathologists³⁵ or validated through immunohistochemistry (IHC) staining.³⁴ It was observed that the predicted TLS spots were highly consistent with these annotated regions (Figures 3I, 3J, and S4C). Collectively, these findings highlight the effectiveness and reliability of SpaLinker in accurately identifying TLS across diverse cancer types.

SpaLinker accurately identifies TNI regions across cancers

The TNI regions, which usually harbor unique cell types^{24,25,36–38} in the front line of the battle between marginal tumor cells and immune cells,³⁹ could be misidentified by methods that

rely on pre-defined tumor areas when tumor cells are distributed diffusely.^{27,40,41} Intuitively, the TNI regions are characterized by an abrupt decrease in tumor cell abundance from the tumor core side to the normal side. Thus, instead of relying on pre-defined tumor regions, SpaLinker determines the TNI regions by detecting the spatial distribution patterns of tumor cells (Figure 4A). In SpaLinker, two functions, *TNIscore* and *DefineTNIregion*, are developed to calculate TNI scores and determine the final TNI regions, respectively (STAR Methods). Furthermore, given the heterogeneity of tumor components within and across tumor samples, we hypothesized that the TNI regions, which encompass distinct domains, may also exhibit heterogeneity. To investigate this point, SpaLinker classifies the TNI regions into different types based on their domain composition (Figures 4A and S5A; STAR Methods).

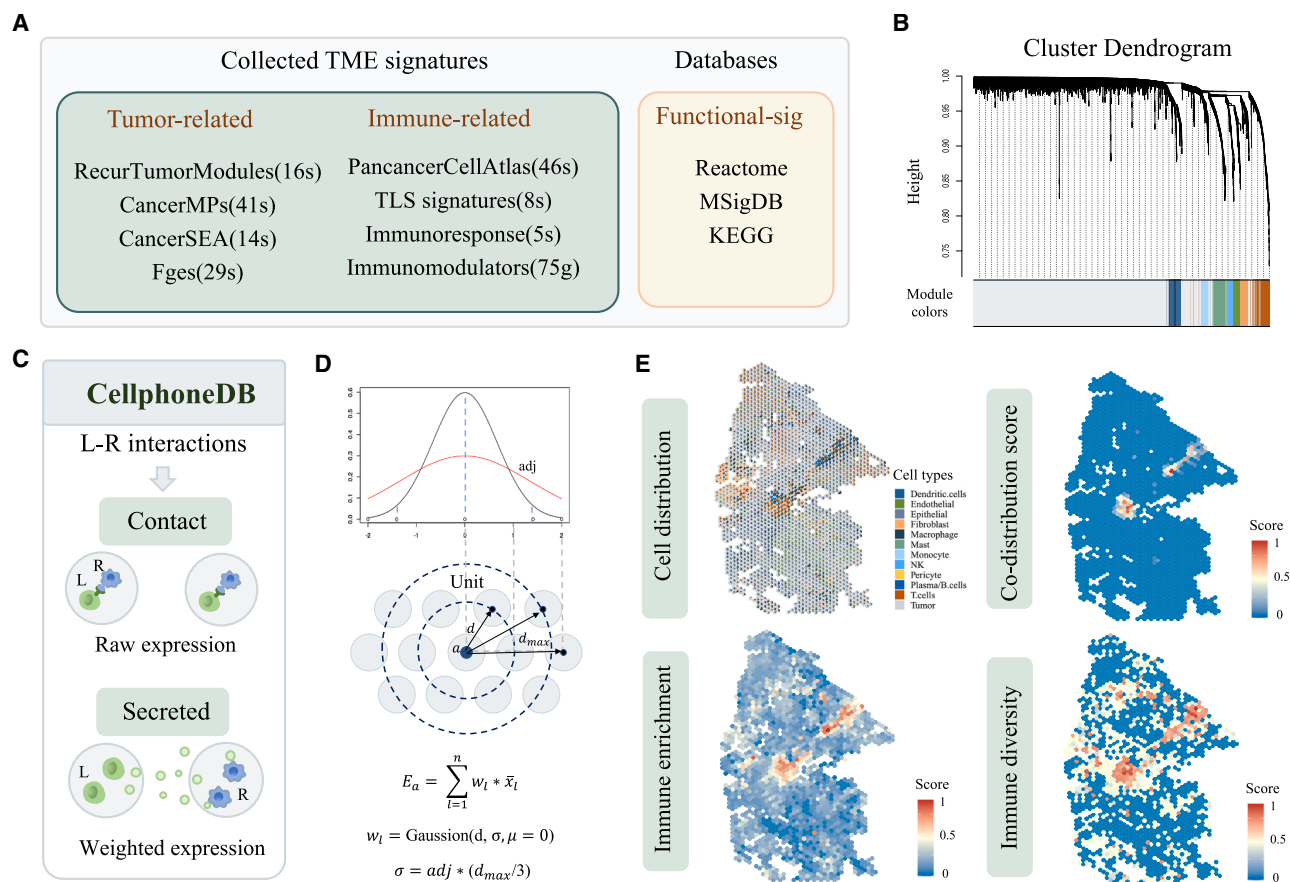


Figure 2. Characterization of the tumor microenvironment with molecular and cellular features

(A) Groups of TME signatures manually collected from literature and functional signatures sourced from databases. The number of corresponding signatures (s) or genes (g) is given in parentheses.

(B) Cluster dendrogram created by WGCNA. Colors represent identified gene coexpression modules.

(C) Illustration depicting ligand-receptor (LR) interactions categorized into contact and secreted types based on their respective interaction distances.

(D) Illustration showing the method to integrate the expression of secreted proteins within the unit (STAR Methods).

(E) Cellular features characterized by SpaLinker.

CancerMPs, cancer meta-programs; CancerSEA, cancer single-cell state atlas; Fges, functional gene expression signatures.

To examine the capability of SpaLinker in identifying TNI regions, we applied it to 10 ST samples across four cancer types. These samples were sourced from either formalin-fixed, paraffin-embedded (FFPE) or frozen tissues and were sequenced by distinct platforms, including two RCC³³ (FFPE, 10× Visium) slices, one squamous cell carcinoma²⁵ (SCC; frozen, Spatial Transcriptomics) slice, one pancreatic ductal adenocarcinoma⁴² (PDAC; frozen, Spatial Transcriptomics) slice, and two HCC²⁴ (frozen, 10× Visium) slices with annotated regions and four HCC³² (frozen, 10× Visium) slices collected from the leading edge of tumor sections. As a result, we detected continuous TNI regions that can effectively distinguish tumor and normal compartments in all samples (Figures 4B and S5B). To confirm these TNI regions, we divided them into tumor-side boundary and normal-side boundary segments (Figures 4C and S5C) and compared their genomic copy number variation (CNV) scores, which are indicative characteristics of malignant cells. Remarkably, we found that the tumor-side boundaries consistently

exhibited significantly higher CNV scores compared to the normal-side boundaries (Figures 4D and S5D), suggesting an aggregation of tumor cells in the tumor side of TNI regions. Furthermore, we observed that the TNI identified by SpaLinker is consistent with the leading edge of tumors when aligning them with hematoxylin and eosin (H&E) images or comparing them with annotations from pathologists (Figures 4E and S5E–S5G). Notably, the cell composition patterns of TNI regions in two HCC slices aligned with the tumor immune barrier (TIB) structures, which are spatial niches located near the tumor boundary and enriched with SPP1⁺ macrophages and cancer-associated fibroblasts (CAFs)²⁴ (Figures 4F and 4G). In addition, we applied our algorithm to a colorectal cancer (CRC)⁴³ slice generated from the Visium HD platform, achieving single-cell scale resolution. We observed that the identified TNI regions were accurately aligned with the edges of annotated tumor regions (Figure 4H).

Furthermore, we compared the TNI regions identified by SpaLinker with the tumor boundaries (Bdy) determined by

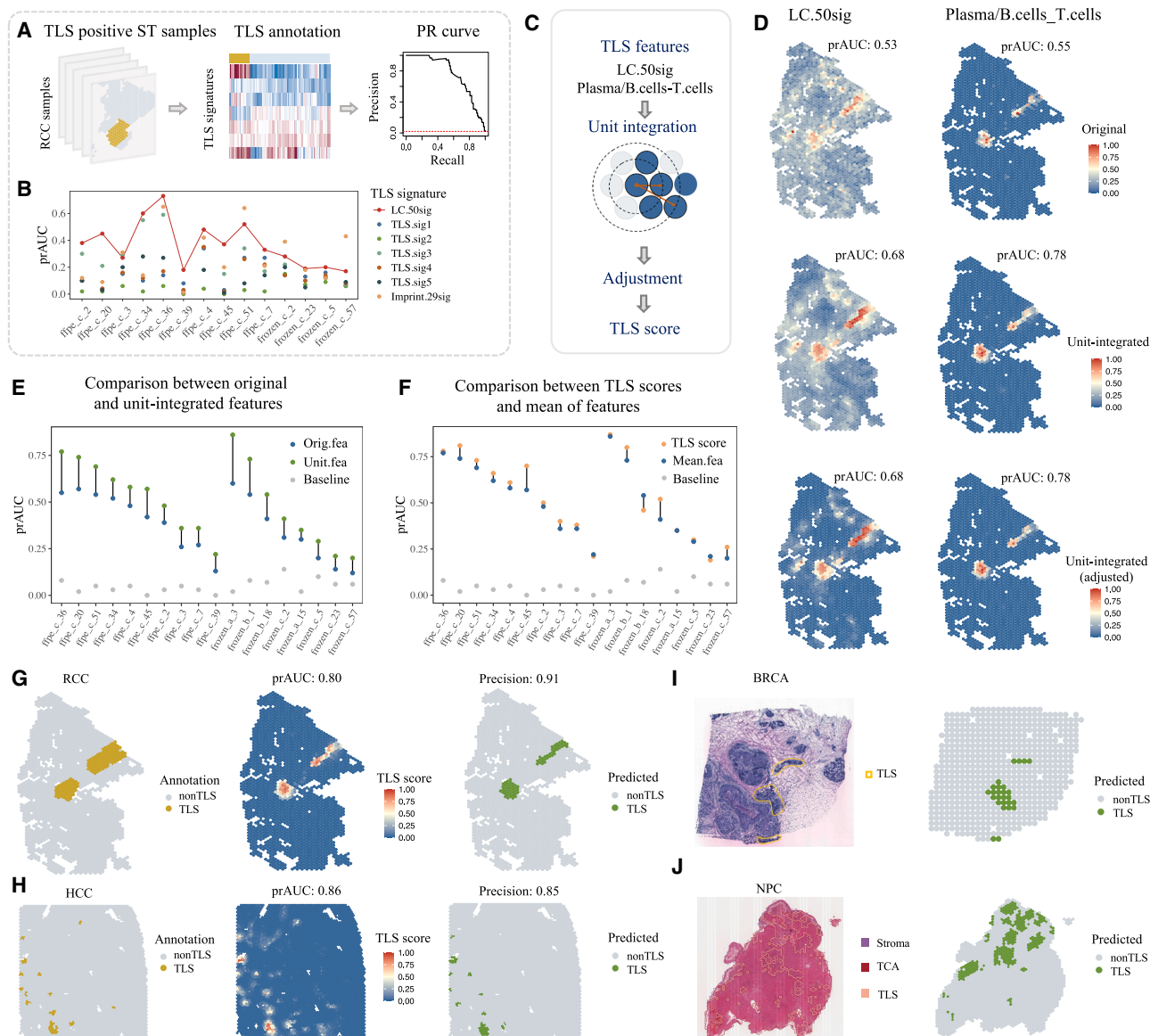


Figure 3. SpaLinker accurately identifies TLSs across cancers

(A) Procedure for comparing the predictive power of multiple TLS signatures using a well-annotated RCC dataset (STAR Methods).

(B) PR-AUC for each TLS signature. The line highlights the LC.50sig, which overall represents the best performance.

(C) Infographic describing the TLS detection strategy. The schematic diagram uses colored spots to represent different domains (STAR Methods).

(D) Visualization of two TLS features, each showing three processed values for comparison.

(E) Dumbbell plot showing the changes in the mean of PR-AUC between the original features (Orig.fea) and the unit-integrated features (Unit.fea).

(F) Dumbbell plot comparing the PR-AUC between final TLS score and mean value of unit-integrated TLS features.

(G and H) Plots showing TLS annotation from pathologists (left), TLS scores (middle), and TLS spots (right) predicted by SpaLinker for one RCC (G) slice and one HCC (H) slice.

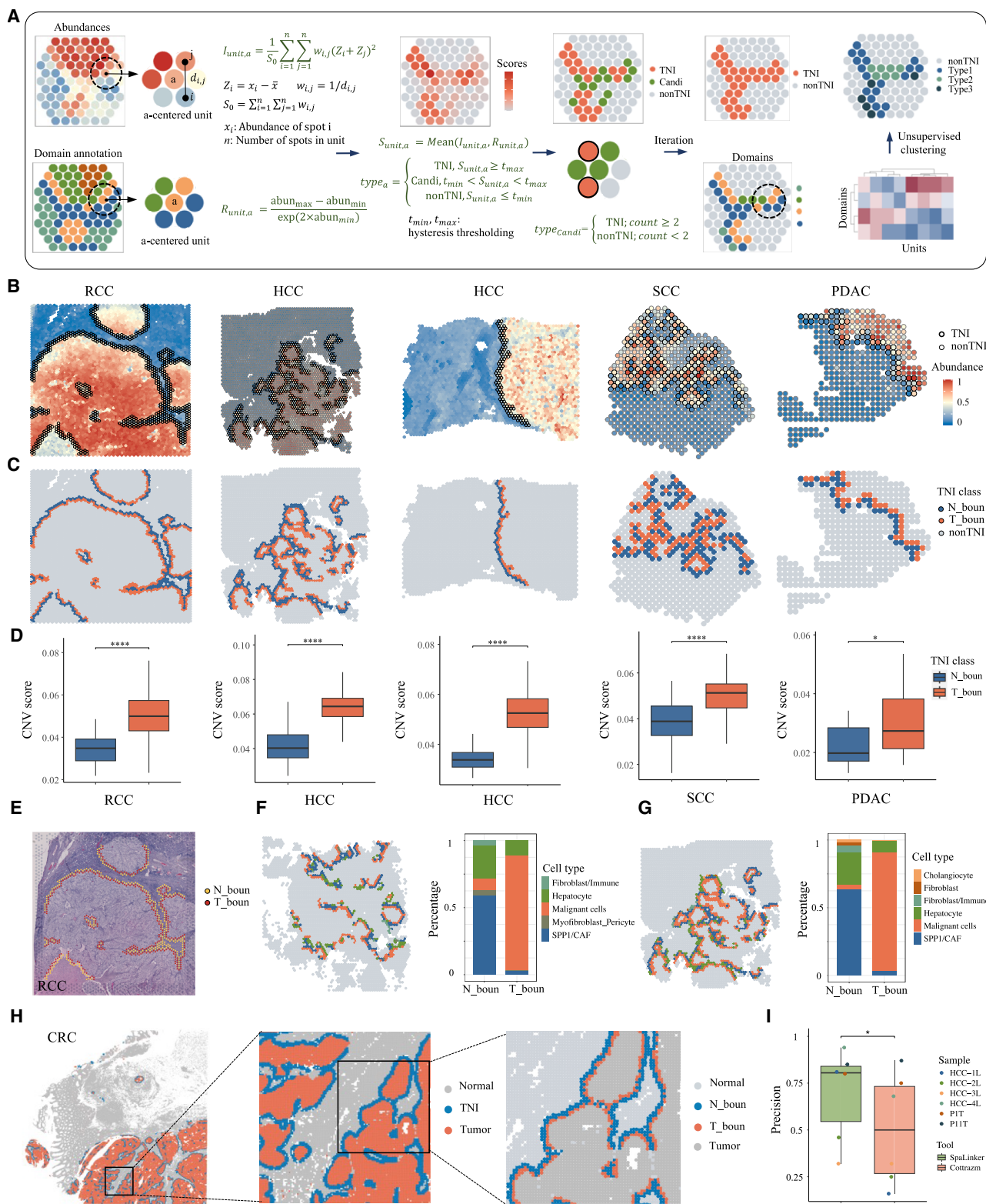
(I) Histologic image (left) marked with TLS regions and identified TLS spots (right) of one BRCA slice.

(J) Histological image (left) and identified TLS spots (left) of one NPC slice. Morphological regions of images were annotated by pathologists into three distinct categories: stromal tissue, TLS, and tumor cell aggregates (TCA).

RCC, renal cell carcinoma; HCC, hepatocellular carcinoma; BRCA, breast cancer; NPC, nasopharyngeal carcinoma.

Cottrazm.⁴⁰ The leading-edge spots of six HCC slices were labeled as ground truth based on tumor tissues annotated by pathologists. Our results indicate that SpaLinker demonstrated more accurate outcomes (Figure 4I). Specifically, SpaLinker

preferentially identified continuous interface regions between tumor and adjacent normal tissue, whereas Cottrazm tended to classify any neighboring spot as Bdy, even when these spots were discrete within the tumor core (Figure S6). Collectively,



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these analyses demonstrate the effectiveness of SpaLinker in accurately capturing the leading-edge interfaces between the tumor and the normal compartments.

SpaLinker links spatially resolved TME features with clinical phenotypes

Given the scarcity of clinical annotations for ST data, further investigation is needed to elucidate the clinical implications of the TME. To this end, SpaLinker designs a phenotype-driven framework that integrates ST data with clinical information obtained from bulk RNA-seq profiles using NMF technology.⁴⁴ Briefly, the process begins with a filtering step to select genes significantly associated with specific phenotypes from annotated bulk RNA-seq profiles. Then an NMF-based analytical strategy is designed to determine multiple phenotype-related factors from filtered expression profiles, which typically represent distinct mechanisms underlying the phenotypes.^{45–48} Utilizing the NMF basis matrix, SpaLinker enables recovery of these phenotype-related factors across ST samples, facilitating comprehensive analyses to widely establish connections between spatial characteristics and patient phenotypes (Figures 5A–5C; STAR Methods).

To demonstrate the effectiveness of SpaLinker in linking phenotypes, we tested its ability to accurately recover phenotype-related factors from two ST cases with ground truth. In the first case, we employed a TLS-labeled ST dataset of RCC patients and simulated 500 pseudo-bulk profiles with varying TLS content, ranging from 0% to 100% (Figures 5D and S7A). By considering the TLS content of the pseudo-bulk data as phenotypes, SpaLinker identified two TLS-related factors. The Gene Ontology (GO) enrichment analysis showed that factor 1, negatively correlated with TLS content, was associated with tumor-related terms, whereas factor 2, which exhibited a strong positive correlation with TLS content, was linked to immune response (Figures S7B and S7C). By recovering these factors in the original ST data and two additional TLS-positive FFPE samples, we consistently observed that factor 1 was predominantly expressed in the non-TLS regions, while factor 2 was significantly enriched in the TLS regions (Figures 5E, 5F, and S7D). In the second case, we tested whether SpaLinker could locate the tumor tissue features from an HCC ST dataset, guided by tumor-related factors derived from a liver HCC (LIHC) cohort from The Cancer Genome Atlas (TCGA).⁴⁹ To evaluate the robustness of the NMF-based strategy with respect to sample size and rank parameter, we generated bulk RNA-seq profiles of varying sizes by randomly resampling the LIHC cohort

(STAR Methods). We then performed NMF decomposition on each resampled dataset, using rank values ranging from 2 to 4. Our results indicate that the NMF clustering remained relatively stable across different ranks and sample sizes, with cophenetic coefficients consistently above 0.9 (Figure 5G). Furthermore, although the overall AUC power (defined as $(\text{AUC} - 0.5) \times 2$) of the factors decreased with smaller sample sizes, the expression patterns of factors recovered by SpaLinker exhibited a strong correlation with tumor cell distribution in four HCC ST slices (Figures 5G–5I, S8A, and S8B). Pathway enrichment analysis revealed that these factors represent diverse components of tumor and normal tissues. For example, when two tumor-related factors were identified, one was related to extracellular matrix (ECM) organization, while the other was associated with pathways involved in cell-cycle regulation (Figure 5J). Collectively, these results demonstrate the capability and robustness of SpaLinker in identifying phenotype-related factors from bulk RNA-seq data and accurately linking them to corresponding spatial characteristics in tumor tissues.

SpaLinker de novo identifies phenotype-associated spatial TME features

SpaLinker identified distinct TNI groups in RCC linked to adverse survival

Previous studies have indicated that the tumor boundary, enriched in epithelial-mesenchymal transition (EMT)-positive RCC cells, adversely affects patient prognosis by promoting tumor invasion and hindering immune infiltration.⁵⁰ However, it remains unclear whether these effects are consistently observed in such regions. To this end, we first explored the diversity of TNI regions using the RCC dataset.³³ SpaLinker detected clear TNI in 22 out of 24 RCC samples (Figures 4B, S5B, and S9), with each being further categorized into distinct TNI types based on the spatial domain composition (Figures S5A and S10). By assessing gene enrichment or depletion in each TNI type, we identified 31 TNI-specific genes predominantly enriched across 54 TNI types in 18 RCC samples. We then categorized these TNI types into four distinct groups (Figure 6A; STAR Methods). Meanwhile, these genes exhibited three patterns, which were related to distinct biological processes and displayed unique enrichment among TNI groups (Figures 6A, 6B, and S11A). Specifically, gene pattern 1, associated with B cell activation, and gene pattern 2, associated with ECM organization, both showed a preferential enrichment in two of the TNI groups, whereas gene pattern 3, related to major histocompatibility complex (MHC), consistently exhibited enrichment in all TNI groups. Moreover,

Figure 4. SpaLinker accurately identifies tumor-normal interface across cancers

- Workflow for the identification and classification of TNI regions (STAR Methods).
- Visualization of TNI spots (circled) showed with tumor abundances in renal cell carcinoma (RCC), hepatocellular carcinoma (HCC), squamous cell carcinoma (SCC), and pancreatic ductal adenocarcinoma (PDAC).
- Visualization of tumor-side boundary (T_boun) and normal-side boundary (N_boun) of TNI regions defined in (B).
- Boxplots displaying the distribution of CNV scores between T_boun and N_boun in (C). Group comparisons were performed using a two-tailed unpaired Wilcoxon rank-sum test (ns, not significant; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$).
- TNI spots aligned with H&E image of an RCC section.
- (F and G) TNI regions of two HCC samples annotated with main cell types obtained from the original study (left). Bar plots showing the distribution of cell types in T_boun and N_boun classes (right).
- Visualization of identified TNI regions in one colorectal cancer (CRC) slice.
- Comparison of SpaLinker and Cottrazm. Group comparisons were performed using a paired Wilcoxon rank-sum test.

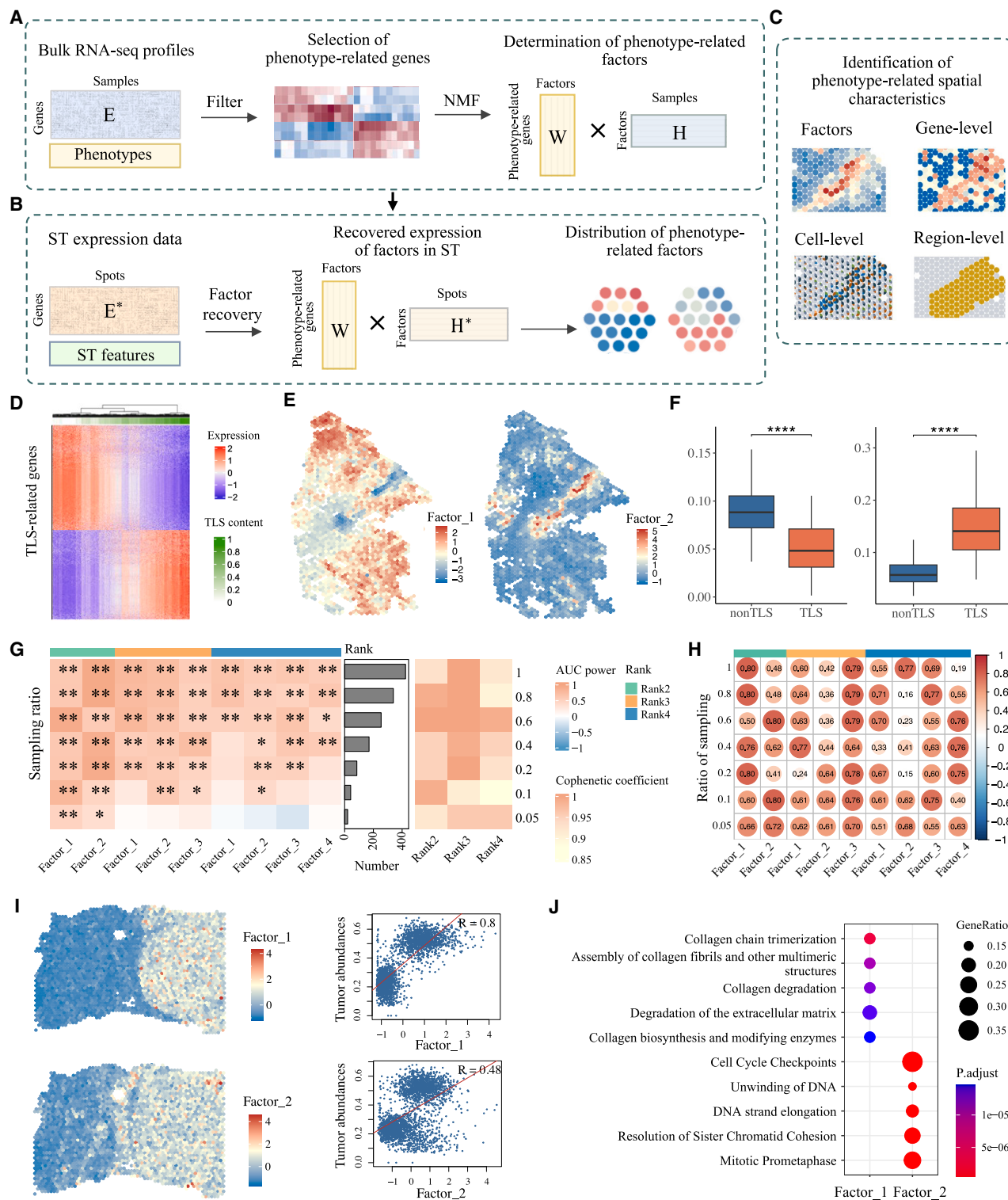


Figure 5. SpaLinker links ST data with clinical phenotypes by integrating bulk and spatial sequencing data

(A and B) Illustration showing the phases of SpaLinker to determine phenotype-related factors (A) and recover them in the ST profiles (B); see [STAR Methods](#). (C) Exploration of the relationship between spatial features of the TME and phenotype-related factors. (D) Heatmap showing genes significantly correlated with TLS content in 500 pseudo-bulk profiles. The gene expression in each pseudo-bulk sample was averaged across spots.

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we noticed that 11 out of 12 genes in pattern 1 were a subset of imprint.29sig, suggesting a potential association between TLS and TNI regions. To further understand their spatial relationship, we considered TNI types to be TLS adjacent if they shared neighboring spots with TLSs (Figure S11B). Interestingly, we found that two TNI groups with high expression of gene pattern 1 significantly enriched TLS-adjacent TNI types (Fisher's exact test with p values 0.02 and 0.00075, respectively; Figure 6A). Finally, based on the enrichment of gene patterns and spatial proximity to TLSs, we annotated the four TNI groups as follows: (1) TLS-adjacent TNI (TLSa), (2) TLS-adjacent and ECM-related TNI (TLSa/ECMr), (3) ECM-related TNI (ECMr), and (4) basic TNI (Basic). This combined analysis revealed a diversity of TNI types in RCC samples, which are intricately shaped by unique spatial characteristics.

To investigate the prognostic impact of these TNI groups, we applied SpaLinker to the TCGA KIRC cohort and determined three factors that were significantly linked to improved survival, as well as seven factors that were related to poor survival (Figures 6C and 6D; Table S3). When recovering these factors in ST samples, we found that both the TLSa and the TLSa/ECMr groups exhibit an enrichment of survival-negative factor 5 (Figure 6E), which was related to immune response pathways, such as inflammasomes, RUNX1 and FOXP3 control of development of regulatory T lymphocytes (Tregs), and neutrophil degranulation (Figure S12A). In general, TLSs are considered to contribute to better clinical outcomes in tumors.^{51,52} However, it is evident that the location of a TLS within the tumor (invasive margin or center) could impact the prognostic role of TLSs in cancers.^{51,53,54} Additionally, further analysis confirmed that both TLS signatures, imprint.29sig and LC.50sig, were linked to shorter survival times in the KIRC cohort (Figures 6F and S12B). These results indicate an unusual impact of TLS on the survival of RCC patients, potentially influenced by the spatial relationship of TLS and TNI.

Remarkably, we observed coenrichment of survival-negative factor 6 and survival-positive factor 1 in two ECM-related TNI groups (Figure 6E). Survival-positive factor 1 was linked to pathways such as deactivation of the beta-catenin transactivating complex and GPCR ligand binding. Conversely, survival-negative factor 6, consistent with the characteristics of ECM-related TNI, was related to ECM organization (Figure S12A), which plays a vital role in tumor development, including proliferation, metastasis, angiogenesis, and immune evasion.⁵⁵ Further investigation revealed that survival-negative factor 6 was specifically expressed in fibroblast cells, while survival-positive factor 1 was primarily expressed by endothelial and pericyte cells (Figure S12C).

Notably, fibroblast cells and endothelial/pericyte cells were coenriched across the two ECM-related TNI groups (Figure 6E). These observations imply the presence of antagonistic components, which have opposite effects on patient outcomes, within the ECM-related TNI region. Collectively, the combined analysis indicates that the TNI in RCC tumors, particularly those adjacent to TLSs or exhibiting high expression of ECM-related genes, are linked to unfavorable prognosis due to various underlying mechanisms.

To further determine the critical spatial features associated with survival-related mechanisms influenced by TNI structures, we subsequently applied SpaLinker to explore the molecular and cellular characteristics within TNI groups. First, we validated the generalizability of the three gene patterns using the TCGA KIRC cohort,⁴⁹ and strong correlations were observed among the genes within each pattern (Figure 6G). We noticed that the widely enriched TNI-specific gene *TMSB4X*, which plays a crucial role in cell proliferation, migration, and differentiation,⁵⁶ is associated with shorter survival time in KIRC and LIHC patients (Figure 6H). We further localized the distribution of this gene in the tumor boundaries, normal boundaries, and non-TNI regions. It is evident that *TMSB4X* was significantly up-regulated in the boundary region of RCC samples, particularly in the normal side (Figure 6I). A similar enrichment was observed in HCC samples, although the distinction between tumor and normal boundaries was not always clear (Figure S13). This analysis indicated the role of the gene *TMSB4X* in the TNI structures and suggested its potential as a prognostic marker in these cancers. Additionally, the assessment of immune and cancer-related features demonstrated biological relevance aligned with gene patterns (Figure S14A). For instance, the CAF signature, matrix signature, and EMT-related programs (MP12, MP13, and MP15) exhibited high enrichment in the ECMr and TLSa/ECMr groups, which aligns with a previous study that reported an enrichment of high EMT-expressing RCC cells at the TNI.⁵⁰ When focusing on the cellular characteristics of TNI regions, we observed that, as expected, the TNI types within TLS-related groups exhibited significant enrichment of immune cells, despite these cells being typically excluded from the tumor cores (Figure 6J). Interestingly, we found that fibroblast cells and tumor-associated macrophages (TAMs) were preferentially codistributed in the ECM-related TNI regions (Figures 6K, 6L, and S14B), which exhibited high expression of the TNI-specific gene fibronectin 1 (*FN1*). The FN1⁺ TAMs have been previously reported as a specific pro-tumor macrophage subset in kidney cancer.^{50,57} Collectively, these results suggest that the fibroblasts and TAMs possibly

(E) Expression of predicted factor 1 and factor 2 in a renal cell carcinoma (RCC) case (frozen_b_1).

(F) Boxplots showing the distribution differences of two factors in (E) between TLS and non-TLS regions. The group comparisons were performed using a two-tailed unpaired Wilcoxon rank-sum test (ns, not significant; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$).

(G) Robustness of NMF-derived factors across varying sample sizes and rank parameter. Left: heatmap displaying the AUC power for tumor-related factors, with rows representing the subsampling ratio of the LIHC cohort and columns indicating the rank value (2–4). Values with AUC power greater than 0.5 (**) and 0.4 (*) are marked. Center: bar plot showing the corresponding number of samples at each ratio. Right: Heatmap of cophenetic coefficients, reflecting the stability of NMF clustering for each parameter combination.

(H) Pearson correlation between the tumor abundances and the recovered tumor-related factors in one HCC ST slice (HCC-1L).

(I) Visualization of the distribution of two tumor-related factors (rank = 2, ratio = 1) in HCC-1L. Scatterplots showing the Pearson correlation between factors and tumor abundance.

(J) Pathway enrichment of tumor-related factors (rank = 2). The top five pathways with an adjusted p value less than 0.05 are displayed.

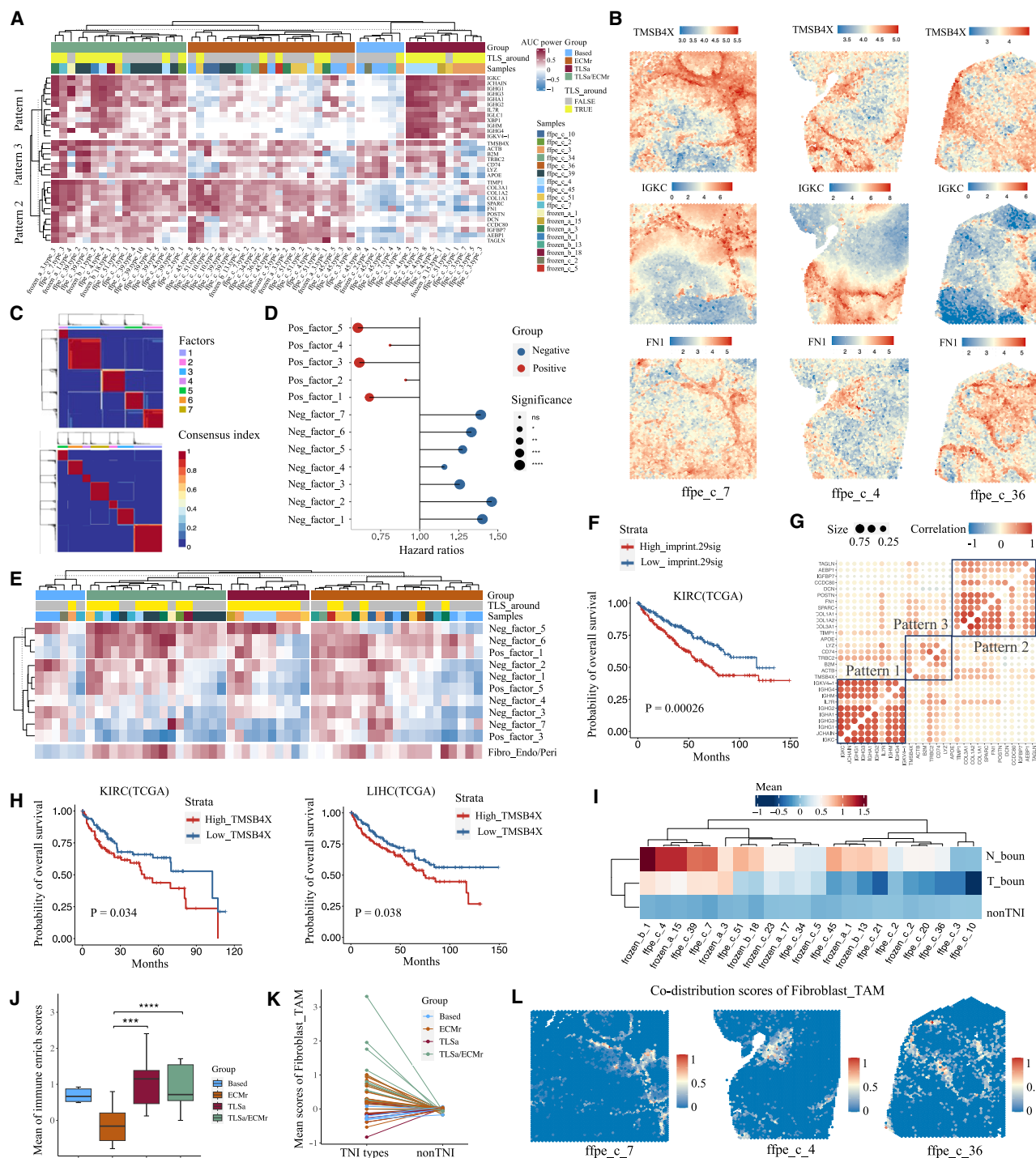


Figure 6. TNI groups of RCC with distinct spatial characteristics and survival associations

(A) Heatmap showing four TNI groups and three gene patterns obtained by unsupervised hierarchical clustering based on AUC powers. Each TNI type is annotated with TNI groups, spatial relationship with TLS (TLS_around), and samples.

(B) Visualization of three cases represented for TLSa/ECMr (left), TLSa (middle), and ECMr (right) groups, respectively. They are colored by three genes, represented by gene pattern 1 (IGKC), pattern 2 (FN1), and pattern 3 (TMSB4X), respectively.

(C) NMF consensus matrices obtained from survival-positive (top) and -negative (bottom) gene groups in the KIRC cohort (TCGA).

(D) Hazards ratios of survival-positive and survival-negative factors in the KIRC cohort (TCGA). Points are colored by factor group, and the size of points represents significance of p value (ns, not significant; $p < 0.05$; $p < 0.01$; $p < 0.001$; and $p < 0.0001$).

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jointly contribute to the ECM remodeling in the interface regions and are associated with shorter survival of RCC patients.

SpaLinker reveals the prognostic significance of heterogeneous components of the TME in HCC

Liver cancer is one of the deadliest types of tumors, with HCC being the predominant histological subtype.⁵⁸ Previous studies have described the heterogeneity of HCC and its clinical significance at low spatial resolution.^{59,60} In this study, we applied SpaLinker to an ST dataset of HCC²⁴ to gain insights into the prognostic relevance of the TME components. First, we conducted spatial domain identification on seven HCC slices and labeled them with the most enriched cell types (Figure S15A). To elucidate the association between TME heterogeneity and patient prognosis, SpaLinker identified three survival-positive and three survival-negative factors from the TCGA LIHC cohort, all of which had a significant impact on patient outcomes (Figures 7A and S15B; Table S3). We then assessed the distribution of these factors in ST data and performed unsupervised clustering on domains that exhibited enrichment (AUC power >0.4) for at least one factor. As a result, we identified three distinct domain clusters across the HCC samples, each exhibiting a specific pattern of factor enrichment. Interestingly, we discovered that these clusters closely align with the cell annotation classification (Figure 7B). Therefore, we categorized these clusters into three groups: the hepatocyte group, the SPP1/CAF group, and the malignant group.

Consistent with prior knowledge, the hepatocyte group showed a significant positive correlation with survival-positive factors, whereas the malignant group exhibited enrichment of all survival-negative factors. However, the SPP1/CAF group also demonstrated a notable enrichment of two survival-negative factors (Figure 7B), suggesting their unfavorable impact on patient outcomes. This result is consistent with the finding of the original study, which reported an association between the TIB, composed of SPP1⁺ macrophages and CAFs, and poor outcomes.²⁴ As described above, the SPP1/CAF spots are mainly located within the TNI regions of HCC slices. To further investigate the potential molecular mechanisms associated with prognosis, we analyzed the interactions of LR pairs within the SPP1/CAF group using SpaLinker. This analysis revealed recurrent LR interactions within the SPP1/CAF regions, including the *CD74-MIF/COPA* pairs and interactions related to the tumor necrosis factor receptor superfamily (*TNFRSF*) (Figure 7C). Notably, the interaction scores for *CD74-MIF* were significantly enriched in the SPP1/CAF compartments across

multiple samples, except for sample P5T, which contained only a few SPP1/CAF spots (Figures 7C–7E). Recent studies have highlighted that the macrophage migration inhibitory factor (*MIF*) is the upstream regulator of *SPP1* and promotes migration in TAMs,⁶¹ while our results suggest that it may play this role via interaction within TIB structures.

Furthermore, we noticed that not all the domains primarily composed of malignant cells were linked with adverse prognosis (Figure S15C), indicating the presence of heterogeneity within tumor cores. To further investigate the characteristics of malignant domains affecting prognosis, we examined the differences in activities of tumor cell state programs and hallmark pathways between malignant domains within the malignant group and those that were not. Our analysis revealed that domains in the malignant group always exhibited higher expression of several meta-programs associated with worse outcomes,⁴⁶ including cell cycle (MP1, MP2, and MP3), MYC (MP20), respiration (MP21) programs, and proteasomal degradation (MP8) (Figures 7F and S15D). Likewise, sustained high activities of pathways related to tumor growth, metabolism, and proliferation were observed within the malignant group, such as E2F targets, G2M checkpoint, and MTORC1 signaling (Figure S15E). Collectively, these results highlight the tumor state heterogeneity within and between HCC tumors, with more active malignant domains often being associated with poorer prognosis.

SpaLinker identified spatial regions related to immunotherapy response in melanoma

The benefits of immune checkpoint therapy are limited to a subset of patients due to the complexities of tumor heterogeneity. In this study, we applied SpaLinker to investigate TME features that underlie the immune checkpoint blockade (ICB) response in melanoma patients. We analyzed an ST dataset of melanoma patients diagnosed with stage III disease.⁶² First, based on pathologic annotation, we labeled the identified domains of the two samples as melanoma, stroma, and lymphoid regions (Figures 7G and S16A). Using 98 bulk samples⁶³ treated with anti-PD-1 therapy as a guide, SpaLinker determined two NMF factors that showed a significant positive correlation with the ICB response, as well as two factors that demonstrated a significant negative correlation (Figure S16B; Table S3). By recovering the relative expression of each factor in ST profiles, we found that two negative factors associated with ICB non-response patients, linked to cellular autophagy and developmental processes, are primarily localized in melanoma regions. In contrast, positive factor 2, related to immune response and

(E) Heatmap showing the AUC powers of survival-related factors. Columns were split by TNI group, and rows were split using unsupervised hierarchical clustering. The single heatmap at the bottom is the AUC power of codistribution scores between fibroblast (Fibro) and endothelial/pericyte (Endo/Peri) cells.

(F) Kaplan-Meier plot showing differences in overall survival between patients with high and low scores of imprint.29sig. The statistical significance was calculated by a two-tailed log-rank test.

(G) Pearson correlation of expression between three gene patterns in the KIRC cohort from TCGA. The size of the point represents the absolute value of the correlation coefficient.

(H) Kaplan-Meier plot showing differences in overall survival between patients with high and low levels of the *TMSB4X* gene in the KIRC and LIHC cohorts.

(I) Heatmap showing mean expression of gene *TMSB4X* in tumor boundary (T_boun), normal boundary (N_boun), and non-TNI regions (nonTNI). Only samples included in four TNI groups are displayed, and the expression of *TMSB4X* in each sample was converted to the Z score.

(J) Boxplot showing the comparison of mean immune enrichment scores across four TNI groups. Group comparisons were performed using a two-tailed unpaired Wilcoxon rank-sum test.

(K) Comparison of codistribution of fibroblasts and TAMs between each TNI type and non-TNI regions.

(L) Visualization of codistribution scores for fibroblasts and TAMs in three representative cases.

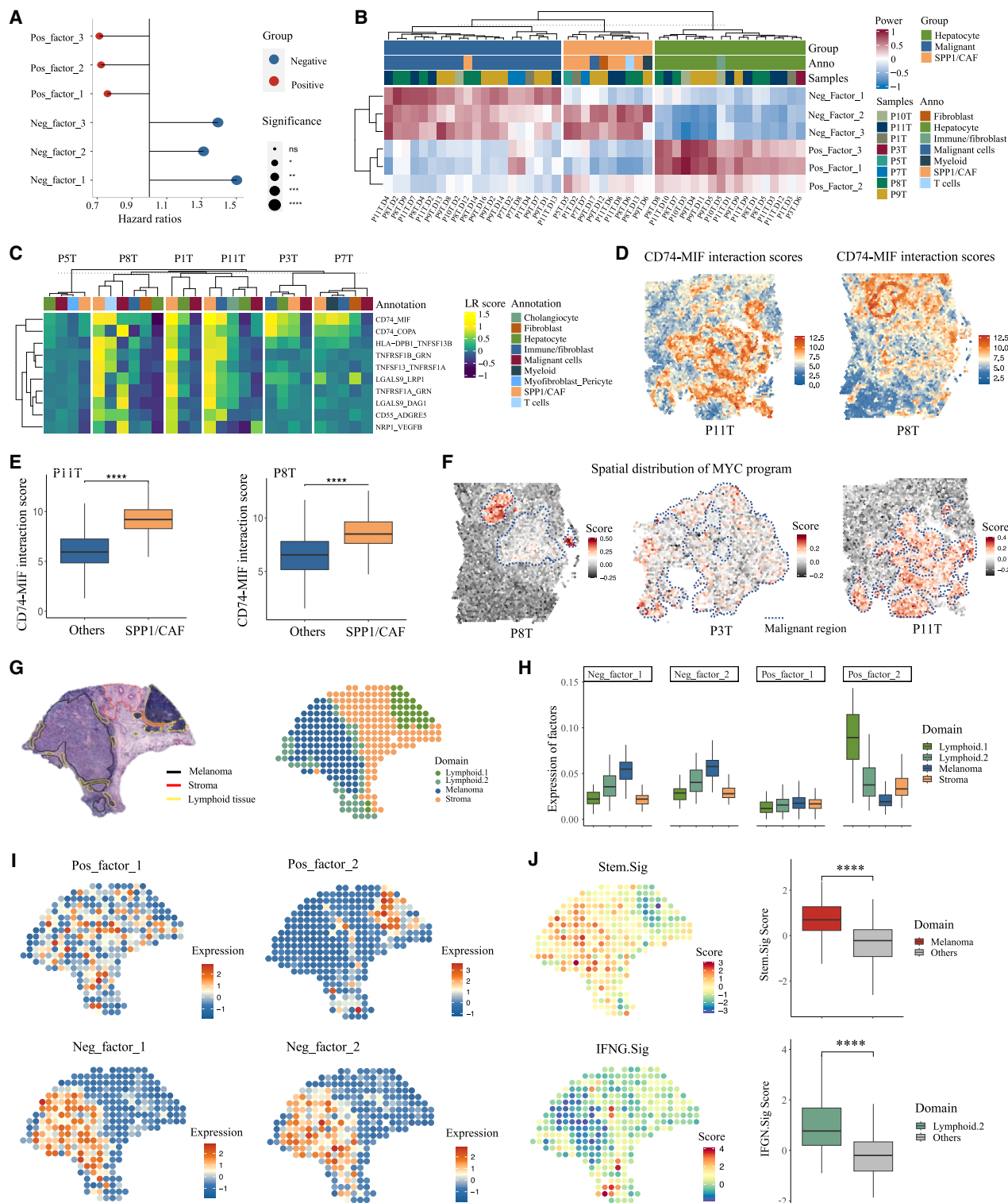


Figure 7. The spatial TME components in HCC and their prognostic impact

(A) Hazards ratios of survival-positive and survival-negative factors in the LIHC cohort (TCGA). Points are colored by factor group and the size of points represents significance of p values (ns, not significant; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$).

(B) Heatmap showing the enrichment or depletion of factors across HCC domains. Groups were obtained by unsupervised hierarchical clustering.

(legend continued on next page)

regulatory pathways, exhibited high enrichment in lymphoid regions (Figures 7H, 7I, and S16C–S16E). As is well known, PD-1 signaling plays an important role in the induction and maintenance of immune tolerance within the TME.⁶⁴ Therefore, the disruption of the PD-1 signaling pathway, guided by anti-PD-1 therapy, may lead to an enhanced immune response, making this pathway a promising target for cancer immunotherapy.

To further confirm the spatial regions related to anti-PD-1 therapy response, we examined well-established predictive gene signatures collected within SpaLinker. We found that the Stem.sig,⁶⁵ which reflects the level of cancer stemness and is associated with ICB resistance, was significantly enriched in the melanoma regions, aligning with the response-negative factors (Figure 7J). Notably, the IFNG.sig⁶⁶ exhibited specifically expression in the peritumoral lymphoid region (Figure 7J). Active interferon- γ (IFN- γ) signaling has been demonstrated as one of the biological features responsive to PD-1 checkpoint blockade.⁶⁶ Collectively, our results indicate that the immune cell-infiltrated regions in melanoma patients may be associated with the ICB response influenced by more complex factors, such as localization and functional signaling.

DISCUSSION

Advances in scRNA-seq and ST technology have opened a new avenue for uncovering the heterogeneity of the TME and its complex interplay with clinical outcomes.^{23,36,67,68} However, the critical challenge lies in systematically analyzing the accumulating ST data, which often lack patient-level annotations, to elucidate their implications for clinical outcomes. In this study, we propose SpaLinker, an innovative analytical framework, to *de novo* identify phenotype-associated spatial TME features by integrating bulk and spatial sequencing data.

Accumulating evidence underscores the importance of spatial features of the TME in tumorigenesis and patient outcomes. By utilizing diverse ST datasets and corresponding bulk profiles, we showcased the capacity of SpaLinker to effectively characterize the spatial characteristics of the TME and link them with patient phenotypes through an NMF-based strategy. Distinguished from traditional feature-driven methods, which link clinical information by assessing the expression of ST markers in bulk profiles, our NMF-based framework offers several key advantages. Specifically, it quantitatively recovers phenotype-related factors across entire ST regions, allowing for an intuitive exploration of relationships between phenotypes and all spatial features. This addresses a critical limitation of traditional methods, which often fail to accurately capture

unique spatial information, such as cell-cell interactions and spatial locations of structures. Moreover, SpaLinker emphasizes the complex mechanisms underlying clinical phenotypes, even in complex situations like the cooccurrence of opposing factors within similar regions. The consistent recoverability of NMF factors across all ST samples also makes our framework particularly well suited to comparative studies involving multiple samples or different cancer types.

SpaLinker enables *de novo* identification of clinical phenotype-associated spatial TME features with clinical insight. The application to RCC, HCC, and melanoma datasets elucidated novel TME features, including prognostic markers and survival-associated spatial domains, and revealed complex, opposing factors influencing clinical outcomes. These insights emphasize the value of SpaLinker as a pioneering tool to identify clinical phenotype-related spatial features across cancer types. Overall, SpaLinker deepens our comprehension of the complex relationship between spatial organization of the TME and clinical outcomes, while also holding potential for contributing to therapeutic strategies.

Limitations

Despite its potential, SpaLinker faces limitations: (1) for the cellular characterization of TME, it requires input cell annotations from upstream analyses, such as spot deconvolution for low-resolution ST data. (2) The initial version of SpaLinker primarily focuses on ST datasets. As technologies in spatial proteomics and metabolomics technologies^{69–71} continue to advance, we anticipate further development of SpaLinker to integrate these new data types. (3) While we've shown the effectiveness of the NMF-based phenotype-linking strategy with simulated and real phenotypes, challenges arise in determining appropriate NMF factors for phenotypes such as survival time, especially with limited sample sizes. Moreover, integrating TME analysis with clinical phenotypes enhances the interpretability of ST data, but the complexity of the information can complicate results. Future investigations leveraging various AI technologies like the explainable AI (XAI) are expected to address these challenges.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources or questions about code and data should be directed to and will be fulfilled by the lead contact, Qi Liu (qiliu@tongji.edu.cn).

Materials availability

This study did not generate new unique reagents.

(C) Heatmap displaying 10 LR pairs that are recurrent across HCC patients with SPP1/CAF regions. The interaction scores of each LR pair have been scaled within samples.

(D) The distribution of interaction scores of CD74-MIF in two HCC slices.

(E) Boxplots showing the significantly higher interaction score of CD74-MIF in SPP1/CAF regions of HCC slices. Group comparisons were performed using a two-tailed unpaired Wilcoxon rank-sum test.

(F) Spatial plots showing the different distributions of the MYC program in HCC slices.

(G) H&E-stained tissue image with pathologist annotations (left) and assigned domain labels (right) of a melanoma ST slice.

(H) Boxplots showing the relative expression of NMF factors across domains.

(I) Visualization of the spatial distribution of each factor.

(J) Distribution of Stem.sig and IFNG.sig in a melanoma section. Boxplots showing the significant differences of signatures between melanoma/lymphoid.2 domain and other regions. Group comparisons were performed using a two-tailed unpaired Wilcoxon rank-sum test.

Data and code availability

This paper analyzes publicly available data. The sources for the datasets are provided in the [key resources table](#). The region annotations of the HCC, BRCA, and CRC datasets were obtained from the original study. The TLS annotations for the RCC dataset were provided by the author.

The open source SpaLinker R package and tutorial are available at GitHub: <https://github.com/bm2-lab/SpaLinker> and Zenodo: <https://doi.org/10.5281/zenodo.15347554>.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (grant nos. T24250193 and 32341008), Shanghai Pilot Program for Basic Research, Shanghai Science and Technology Innovation Action Plan-Key Specialization in Computational Biology, Shanghai Shuguang Scholars Project, Shanghai Excellent Academic Leader Project, Shanghai Municipal Science and Technology Major Project (grant no. 2021SHZDX0100), and Fundamental Research Funds for the Central Universities.

AUTHOR CONTRIBUTIONS

Q.L., Z.Z., X.C., and C.T. designed the framework of this work. X.C., C.T., K.D., Y.Y., X.Z., B.D., and S.C. performed the analyses. X.C., C.T., Z.Z., G.C., and Q.L. wrote the manuscript with the help of other authors. All authors read and approved the final manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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 - SpaLinker identifies spatial architecture levels of TME features
 - SpaLinker links spatial tumor microenvironment features with clinical phenotypes by integrating bulk and spatial sequencing data
 - Data visualization and survival analysis
- **QUANTIFICATION AND STATISTICAL ANALYSIS**

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.xgen.2025.100893>.

Received: November 28, 2024

Revised: March 28, 2025

Accepted: May 6, 2025

Published: June 5, 2025

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
KIRC bulk RNA-seq	UCSC Xena	https://xena.ucsc.edu/
LIHC bulk RNA-seq	UCSC Xena	https://xena.ucsc.edu/
Melanoma bulk RNA-seq	TIGER	http://tiger.canceromics.org/#/
RCC ST	Meylan et al. ³³	GEO: GSE175540
RCC scRNA-seq	Li et al. ⁵⁰	Mendeley Data: https://data.mendeley.com/datasets/g67bkbnhhg/1
BRCA ST	Andersson et al. ³⁴	EGA: EGAD00001008031
BRCA scRNA-seq	Wu et al. ⁷²	GEO: GSE176078
HCC ST	Wu et al. ³²	GSA: HRA000437
HCC ST	Liu et al. ²⁴	Mendeley Data: https://data.mendeley.com/datasets/skrx2fz79n/1
HCC scRNA-seq	Liu et al. ²⁴	Mendeley Data: https://data.mendeley.com/datasets/skrx2fz79n/1
NSC ST	Liu et al. ³⁵	GEO: GSE206245
NSC scRNA-seq	Liu et al. ³⁵	GEO: GSE162025
PDAC ST	Moncada et al. ⁴²	GEO: GSE111672
PDAC scRNA-seq	Moncada et al. ⁴²	GEO: GSE111672
SCC ST	Ji et al. ²⁵	GEO: GSE144240
SCC scRNA-seq	Ji et al. ²⁵	GEO: GSE144240
Melanoma ST	Thrane et al. ⁶²	https://www.spatialresearch.org/resources-published-datasets/doi-10-1158-0008-5472-can-18-0747/
CRC ST	10X Genomics	https://www.10xgenomics.com/products/visium-hd-spatial-gene-expression/dataset-human-crc
Software and algorithms		
SpaLinker	This work	GitHub: https://github.com/bm2-lab/SpaLinker Zenodo: https://doi.org/10.5281/zenodo.15347554
Seurat v4.3.0	Hao et al. ⁷³	https://satijalab.org/seurat/
AUCell v1.22.0	Aibar et al. ⁷⁴	https://github.com/aertslab/AUCell
GSVA v1.46.0	Hanzelmann et al. ⁷⁵	https://github.com/rcastelo/GSVA
Ucell v2.5.1	Andreatta et al. ⁷⁶	https://github.com/carmonalab/UCell
Cell2location v0.1.3	Kleshchevnikov et al. ⁸	https://github.com/BayraktarLab/cell2location
WGCNA v1.72-1	Langfelder et al. ³⁰	https://fuzzytelin.github.io/bioanth-stats/module-F21-Group1/module-F21-Group1.html
fpc v2.2.13	Hennig et al. ⁷⁷	https://github.com/cran/fpc/tree/master
CopyKAT v1.1.0	Gao et al. ⁷⁸	https://github.com/navinlabcode/copykat
BayesSpace v1.10.1	Zhao et al. ¹⁵	https://github.com/edward130603/BayesSpace
ComplexHeatmap v2.14.0	Gu et al. ⁷⁹	https://github.com/jokergoo/ComplexHeatmap
DESeq2 v1.40.2	Love et al. ⁸⁰	https://github.com/thelovelab/DESeq2
clusterProfiler v4.6.2	Yu et al. ⁸¹	https://github.com/YuLab-SMU/clusterProfiler
ggplot2 v3.4.0	Wickham et al. ⁸²	https://github.com/tidyverse/ggplot2
survival v3.7.0	Therneau et al. ⁸³	https://github.com/therneau/survival
survminer v0.4.9	Kassambara et al. ⁸⁴	https://github.com/kassambara/survminer
NMF v0.28	Gaujoux et al. ⁸⁵	https://github.com/renozao/NMF

METHOD DETAILS

The framework of SpaLinker

SpaLinker is an integrative analytical framework designed for linking spatially resolved TME features with clinical phenotypes by integrating bulk and spatial sequencing data. It contains three functional parts: (1) Characterization of the TME with molecular and cellular

features; (2) Recognition of spatial architecture features; and (3) Linking with clinical phenotypes. To prepare for subsequent functional analysis parts, SpaLinker offers initial steps, including checking the coordinate information and processing expression data, similar to scRNA-seq analysis. The expression matrix and coordination information are the minimum inputs required for SpaLinker to perform basic molecule-level analysis. Additional inputs may be necessary for a more comprehensive characterization of the TME. This can include cell annotations of spots for exploration depending on cellular features, or bulk expression profiles with clinical annotations for linking with phenotype. The STFeature object, built on the flexible S4 object system in R, is designed to efficiently store multi-level features for quick and convenient analysis in SpaLinker. Additionally, SpaLinker supports the input of alternative annotations from other tools, such as clustering and enrichment results, enhancing its flexibility and practicality for a variety of analysis needs.

Data preprocessing

Data preprocessing of spatial transcriptomics

The raw expression matrix and spatial coordinates for each sample were imported into a Seurat object using the R package Seurat⁷³ (Version 4.3.0). For ST data acquired from 10x Genomics Visium platform, SpaLinker utilizes the *STcoordCheck* function to verify and adjust the coordinates of spots, ensuring their proper arrangement on hexagonal lattices. Quality control, sample normalization, dimensionality reduction, and Louvain clustering were conducted using the *SePreprocess* function, following the introduction to analyze spatially-resolved RNA-seq data (Analysis, visualization, and integration of spatial datasets with Seurat · Seurat (satijalab.org)).

Data preprocessing of scRNA-seq references

For RCC single cell dataset from Ruoyan et al (2022),⁵⁰ only single cells from the 10 ccRCC patients were used and cells which annotated as “Low quality”, “Unknown”, “CD4+T_Naive/CM-patient-specific.1”, and “CD4+T_Naive/CM-patient-specific.2” were removed. The remaining cells were classified into 103 subtypes by the original study. For the convenience of subsequent analysis, these cell subtypes were merged into 12 main cell types: Plasma/B cells, T cells, Endothelial, Pericyte, Epithelial, Fibroblast, Monocyte, Dendritic, Macrophage, Mast, NK, and Tumor cells.

Deconvolution of spot-level spatial transcriptomics

Specifically, SpaLinker accepts cell quantifications annotated from various deconvolution tools.^{8,9,11,14} In this study, we selected Cell2location⁸ to integrate ST data and single-cell RNA-seq reference as it has been demonstrated to have well-accepted performance.¹³ Cell2location provides estimated abundance for each cell type, which can be converted to proportions by dividing the sum of each spot. For RCC ST dataset,³³ scRNA-seq data from 10 RCC patients were used as reference,⁵⁰ and the merged cell types of single cells described above were used as the annotation. For HCC, ICC and cHC cases,^{24,32} scRNA-seq reference with cell annotation from one original study²⁴ was used. For the BRCA ST dataset,³⁴ pre-treatment tumors from five patients with HER2+ BRCA were selected as reference.⁷² For nasopharyngeal carcinoma datasets, scRNA-seq reference and main cell types annotated from one NPC patient (GSM4929846) was used.⁸⁶ For the SCC²⁵ and PDAC⁴² ST datasets, paired scRNA-seq references from original studies were used. In detail, the single-cell regression model was trained with default parameters and the Cell2location model was obtained with the parameter *N_cells_per_location*=20, *detection_alpha*=20 for all ST datasets.

SpaLinker identifies molecular and cellular levels of TME features

Collection and assessment of annotated functional gene sets

The tumor microenvironment is an intricate system shaped by the long-term interaction of various states of tumor cells and immune cells. To provide a more intuitive characterization of the TME, we utilized published literature to create a collection of TME-related signatures that represents the major cell states or functional processes. The TME-related signatures were classified into two categories, namely the tumor-related and immune-related gene sets. The tumor-related gene sets were collected from four separate studies, three of which capture recurrent tumor cell states or programs across various cancer types, including RecurTumorModules⁴⁷, CancerMPs⁴⁶, and CancerSEA⁸⁷. The fourth is Fegs,⁸⁸ describing various components of the TME. On the other hand, the immune-related signatures encompass subpopulations of immune cells, markers of TLS and evaluations of responses to immunotherapy, namely PancancerCellAtlas,⁸⁹ TLS signatures^{33,53,90–93} and Immunoresponse^{65,66,94–96} respectively (Figure 2A and Table S1). Additionally, a list of 78 Immunomodulators,⁹⁷ which play a critical role in cancer immunotherapy, were also included in the immune-related features. Moreover, SpaLinker allows exploration of molecular signatures from widely used databases, such as Reactome,⁹⁸ KEGG⁹⁹ and Molecular Signatures Database (MSigDB),¹⁰⁰ thereby enriching the analytical resources available for comprehensive TME analysis.

Specifically, these gene sets can be queried and quantified using the *GetGsetSigScore* function. SpaLinker implements four methods to assess their activities: (1) the *AddModuleScore* function in Seurat⁷³ package (Version 4.3.0); (2) the *AUCell* function in AUCell⁷⁴ package (Version 1.22.0); (3) the *gsva* function in GSVA⁷⁵ package (Version 1.46.0); and (4) the *AddModuleScore_UCell* function in UCell⁷⁶ package (Version 2.5.1). In this study, expression scores of all gene sets for each sample were calculated using *AddModuleScore* function with default parameters.

Determination of gene co-expression modules

SpaLinker implements the R package WGCNA³⁰ (Version 1.72-1) to identify gene co-expression modules. To analysis the relative expression of modules across regions, module eigengenes, which defined as the first principal component of the expression of corresponding modules, were calculated using *moduleEigengenes* function in WGCNA package.

Quantification of LR interaction scores

Cell-cell communication mediated by ligand-receptor complexes is fundamental to diverse biological processes. SpaLinker enables quantification of 1396 LR interactions collected from the CellPhoneDB³¹ database, which includes multi-subunit information of protein complexes.

For ST data, to facilitate the downstream analysis of exploration between spatial features, SpaLinker quantifies the interaction scores of LR pairs for each spot. First, the expression of multi-subunit protein complexes is represented by the minimum values of the members. Then, for each ligand and receptor, the effective expression levels are calculated based on their interaction distance, which is limited by the protein types.^{101,102} Specifically, for membrane proteins, the effective expression is equal to the original expression of spots, since they can only interact with partners which come from the same or contact cells. For secreted proteins, there is a potential for them to spread out and interact with surrounding partners, but the probability of interaction decreases as the distance increases. Therefore, at each spot location, the effective expressions of secreted proteins that can contribute to interactions are determined by two components: the retained intrinsic expression at the spot itself and the contribution from the surrounding spots. Based on this, the unit, which consists of one central spot and its neighboring spots up to n -layers, is defined. The effective expression of the secreted protein at each spot position is calculated as the distance-weighted sum of all expressions within that unit (Figure 2D):

$$E_{i,a} = \sum_{l=1}^n w_l * \bar{x}_{i,l} \quad (1)$$

where:

$$w_l = \text{Gaussian}(d, \sigma, \mu) \quad (2)$$

$$\sigma = \text{adj} * \left(\frac{d_{\max}}{3} \right) \quad (3)$$

$$\sum_{l=1}^n w_l = 1 \quad (4)$$

$E_{i,a}$ is the integrated expression of the secreted protein i in spot a , $\bar{x}_{i,l}$ is the mean expression of i in the layer l , and n is the number of nearest layers of unit, which is selected by the users. The weight for the layer l , denoted as w_l , can be defined using a Gaussian distribution. By default, the mean (μ) of Gaussian distribution is set to 0, while the standard deviation (σ) is determined based on the distance of the maximum layer ($d_{\max}$) and is adjusted with *adj* parameter. Additionally, the sum of the weight should keep to 1. The final interaction scores are calculated as the product of adjusted expression levels between ligands and receptors. In this study, the $d_{\max}$ was set to 4 for each HCC ST sample.

Calculation of cell co-distribution scores

To describe the spatial co-enrichment and co-depletion patterns of two cell types, SpaLinker calculates the co-distribution score for each spot. The proportion of each cell type, obtained through reference integration, is initially normalized to a range of 0-1 within the sample. The co-distribution score of each cell type pair is then calculated as the product of the normalized proportion at each spot.

Measurement of immune infiltration

The degree of immune infiltration reflects the spatial heterogeneity of immune cells using two measures, i.e., immune enrichment and immune diversity. For each spot, the immune enrichment score is determined as the overall percentage of immune cells present. The immune diversity is assessed using the Shannon entropy¹⁰³ (H), which is calculated as

$$H = - \sum_{i=1}^n p_i \log_2 (p_i) \quad (5)$$

Where p_i is defined as the proportion of immune cell type i among all immune cells in each spot, and n is the number of immune cell types.

SpaLinker identifies spatial architecture levels of TME features

Annotation of spatial domains

Spatial domains are functional structures within tissues that possess unique local features.¹⁰⁴ To enhance the pipeline analysis, SpaLinker implements Seurat and BayesSpace,¹⁵ a computational tool known for its strong performance in clustering analysis of sequencing-based ST data,^{105,106} to detect domains. In addition, SpaLinker also accepts user-provided domain annotations as input. For all ST datasets used in this study, the domain annotation was conducted using the *BayesCluster* function. Specifically,

the *qTune* and *qPlot* functions were employed to determine the appropriate number of domains. For each sample, parameter *q* was selected as the point around the elbow of the line in figure from *qPlot* function, and other default parameters were applied.

Feature selection procedure for TLS identification

Based on the definition, the co-localization of Plasma/B cells and T cells is a fundamental cellular feature of TLS. In addition, TLS signatures are commonly used to represent the molecular characteristics of TLS. To select the most representative and powerful TLS signature, we compared the predictive abilities of seven gene sets previously reported in studies^{32,33,53,90–93} (Table S1). This comparison was conducted using a RCC dataset,³³ which includes 18 TLS-positive samples annotated by pathologists. In this step, we excluded four frozen samples, since the imprint.29sig was derived from these samples. In detail, the enrichment scores of these TLS signatures in TLS-positive RCC samples were calculated using the *AddModuleScore* method in *GetGsetSigScore* function. Then we evaluated and compared the performance of signatures in TLS prediction methods using the Precision-Recall Curve, a metric capable of handling imbalanced datasets. As a result, LC.50sig demonstrated superior performance across most samples (Figure 3B). Therefore, we selected LC.50sig and the co-localization of Plasma/B cells and T cells as key features for identifying TLS.

Stepwise identification of TLS

SpaLinker identifies TLS step by step. First, SpaLinker characterizes each spot by assessing the enrichment of LC.50sig signatures using the *AddModuleScore* method in *GetGsetSigScore* function, while also calculating the co-distribution scores of Plasma/B cells and T cells. Next, given the heterogeneity within TLS caused by non-uniform cell distribution, SpaLinker integrates information from units composed of neighboring spots within the same domain to capture more comprehensive characteristics. Specifically, the integrated unit scores are calculated using the same method designed for secreted proteins, with the unit comprising a central spot and its outer layers (one layer for Visium and two layers for ST) within the same domain. Subsequently, to reduce noise from lower signals, the integrated unit scores are adjusted based on significance *p*, which is calculated using a randomized permutation test. For each sample, 1,000 randomized permutations of spot positions (without domain labels) are performed to generate a null distribution. The significance *p* value for each unit score is calculated as the proportion of randomized permutations where the score exceeds the actual unit score. The *p* values are then transformed into $-\log_{10}(p)$ values. Within each sample, the unit scores and transformed *p* values are normalized independently to the range [0, 1]. Then, the normalized unit scores are adjusted by multiplying them with their corresponding normalized $-\log_{10}(p)$ values. This approach down-weights the scores of units that are low and not statistically significant, thereby reducing noise. Finally, a predictive TLS score is calculated as the geometric mean of three adjusted TLS features, resulting in a range of 0 to 1.

To evaluate the improved performance at each step, the values of PR-AUC were evaluated for all processed TLS features, including original features, unit-integrated features and adjusted unit-integrated features. For each TLS-positive sample, the baseline of PR-AUC was determined as the fraction of TLS spots. To identify the final TLS regions in ST samples, we calculated the F1 score values for TLS scores in 18 TLS-positive RCC samples. The F1 score was calculated as the harmonic mean of precision and recall. We used the mean TLS score of 0.2 as the initial threshold, which corresponds to the TLS score associated with the highest F1 score for each sample. For HCC, ICC, and cHC, the threshold was set at 0.4 because the annotated TLS in these samples were more rigorously selected by pathologists to include only spots with germinal centers. Other samples used the default threshold. For ST datasets with ground truth of annotated TLS spots, including RCC, HCC, ICC, and cHC, the precision metric was used to evaluate the quality of the TLS predicted by SpaLinker.

Detection of TNI regions

The TNI regions, consisting of tumor and normal boundaries, are expected to be characterized by units where tumor cells exhibit one-sided clustering patterns. To evaluate the spatial distribution pattern of tumor cells within each unit, SpaLinker designs two measures: (1) A distribution imbalance score, denoted as *I*, is used to assess whether tumor cells are spatially clustered or dispersed by considering both their location and abundances; (2) A difference range score *R*, which measures the data variation, is employed to detect the difference of tumor cell abundances across domains. Specifically, the *I* score of each unit is defined as follows:

$$I_{unit,a} = \frac{1}{S_0} \sum_{i=1}^n \sum_{j=1}^n w_{ij} [(x_i - \bar{x}) + (x_j - \bar{x})]^2 \quad (6)$$

Where $I_{unit,a}$ refers to the *I* score of the unit with spot *a* as its center, *n* is the number of spots in unit, *x* represents the quantified abundances of tumor cells in spot *i* or spot *j*, and $\bar{x}$ denotes the mean abundance of unit. The weight w_{ij} is defined as the inverse Euclidean distance (d_{ij}) between spot *i* and spot *j*:

$$w_{ij} = \begin{cases} d_{ij}^{-1}, & i \neq j \\ 0, & i = j \end{cases} \quad (7)$$

Here, S_0 represents the sum of all w_{ij} :

$$S_0 = \sum_{i=1}^n \sum_{j=1}^n w_{ij} \quad (8)$$

The $I_{unit,a}$ was adjusted by adding a pseudocount of 1 and $\log_2$ -transformed to mitigate the influence of extreme outliers. This calculation method ensures that units with significant differences in tumor abundance and unilaterally clustered distributions

achieve a high I score. Conversely, units with minimal differences in tumor abundance and random spatial distributions receive a low I score.

Typically, due to substantial heterogeneity, the tumor core and adjacent normal regions are segmented into distinct domains during clustering. Thus, the R score for each unit ($R_{unit,a}$) is computed to quantify the range of tumor cell abundance differences across domains. It is defined as follows:

$$R_{unit,a} = \frac{abun_{max} - abun_{min}}{\exp(2 \times abun_{min})} \quad (9)$$

Where $abun_{max}$ and $abun_{min}$ represent the maximum and minimum mean abundances of tumor cells across domains within spot a -centered unit. The numerator captures the absolute difference between maximum and minimum values, while the denominator incorporates the minimum abundance into the adjustment. In this way, units exhibiting significant variations in tumor cell abundance across domains are assigned high R scores, whereas units only contain a single domain receive a score of zero. In cases where cluster information is unavailable, SpaLinker calculate the $abun_{max}$ and $abun_{min}$ by averaging the two highest and two lowest spot values within each unit, respectively. The final TNI score of spot a -centered unit ($S_{unit,a}$) is calculated as the geometric mean of $R_{unit,a}$ and $\log_2$ -transformed $I_{unit,a}$. The resulting TNI scores range from 0 to 1, as the input tumor abundances for each sample are represented as proportions or initially normalized to the range [0, 1].

Next, SpaLinker employs the hysteresis thresholding to iteratively determine the TNI regions. To suppress the noise and ensure the detection of continuous regions, double thresholds, t_{min} and t_{max} , are initially established. The label of spot a ($type_a$) is then determined step by step. Spots with TNI scores below the t_{min} are filtered out and categorized as non-TNI (nonTNI) spots. Conversely, spots with scores above t_{max} are identified as TNI spots, and spots which fall between the t_{min} and t_{max} are labeled as candidate (Candi) spots:

$$type_a = \begin{cases} \text{TNI}, & \text{if } S_{unit,a} \geq t_{max} \\ \text{Candi}, & \text{if } t_{min} < S_{unit,a} < t_{max} \\ \text{nonTNI}, & \text{if } S_{unit,a} \leq t_{min} \end{cases} \quad (10)$$

For each Candi spot, SpaLinker calculates the number of TNI spots in its adjacent layers, denoted as $count$. By default, if two or more TNI spots are present, the Candi spot is relabeled as TNI. Otherwise, it is relabeled as nonTNI:

$$type_{Candi} = \begin{cases} \text{TNI}, & \text{if } count \geq 2 \\ \text{nonTNI}, & \text{if } count < 2 \end{cases} \quad (11)$$

This iterative process continues until the maximum number of iterations (default: 20) is reached or no further changes occur in spot labels. Furthermore, to ensure the detection of connected TNI regions, spatially isolated spots or small regions containing fewer spots than a specified cutoff (default: 10) are filtered out. In SpaLinker, the calculation of interface scores and the determination of TNI spots are implemented using the *TNIscore* and the *DefineTNIregion* functions, respectively.

Determination of TNI types and categories

To differentiate various TNI regions, SpaLinker employs unsupervised clustering to categorize TNI units into distinct types based on their spatial domain composition. For each TNI spot, the corresponding unit is defined by the central spot along with its neighboring TNI spots. If the number of types is not pre-specified by users, SpaLinker employs the R package *fpc*⁷⁷ (Version 2.2.13) to determine the optimal number of clusters. Spatially isolated points are filtered out for each type.

Additionally, TNI spots are further categorized as either tumor or normal boundaries based on the tumor cell density. Within each TNI type, the overall average of tumor cell abundance is calculated first. For each spatial domain within this TNI type, if the average tumor cell abundance is higher than the overall average, the spots within that domain are labeled as tumor-side boundaries. Otherwise, they are labeled as normal-side boundaries.

Identification of TNI regions in spatial transcriptomic datasets

To identify the TNI regions of all utilized ST datasets, we quantified tumor cell abundances using results estimated from Cell2location for all spot-resolution samples. For samples with multiple tumor cell subtypes in the deconvolution, we first calculated the overall tumor cell abundance by summing the contributions from all identified subpopulations. To standardize the TNI scores within the range [0, 1], tumor cell abundances were converted to proportions. Additionally, given the variability in the total number of cells across spots, which potentially affects the accuracy of detecting changes in tumor cell density, we incorporated the information of absolute abundances of tumor cells. Specifically, for each sample, tumor cell abundances of spots were adjusted by adding a pseudo-count of 1, $\log_2$ -transformed to stabilize the variance, and normalized by the sample's maximum value to scale between 0 and 1. The proportion of tumor cells in each spot was then adjusted by multiplying it with the normalized abundance. In this study, the TNI regions for all spot-resolution ST samples were detected using units formed by the nearest one-layer spots. For the CRC slice from the Visium HD platform, data binned at 8 μm were used, and the units were calculated based on the nearest two-layer bins. The tumor abundance of bins annotated as tumor cells was set to 1, while all other bins were set to 0. The parameters t_{min} and t_{max} were set to 0.03 and 0.06 in most samples, with fine-tuning applied to some.

Calculation of CNV scores

The CNV scores of spots were estimated from the spatial transcriptome profiles using the R package CopyKAT⁷⁸ (Version 1.1.0) with default parameters. By inputting the raw count matrix of ST data, CopyKAT estimated the copy number profiles and generated a CNV matrix where rows are spots and columns are 220kb bins in the genomic order. The CNV score of each spot was calculated by averaging the absolute value of copy numbers, which reflects the abundance of tumor cells.

Comparison with Cottrazm

To compare SpaLinker with the Cottrazm⁴⁰ tool, we annotated the leading-edge spots as ground truth based on pathological annotations, utilizing six HCC slices with clear leading edges. First, small tumor or non-tumor regions containing fewer than three spots were excluded from the analysis. Subsequently, neighboring tumor or non-tumor spots were grouped as tumor-normal boundaries. For tumor boundary (Bdy) identification using Cottrazm, spot labels for these slices were downloaded from the SpatialTME¹⁰⁷ database. For each slice, we employed the precision metric to evaluate the quality of the identified TNI and Bdy spots.

SpaLinker links spatial tumor microenvironment features with clinical phenotypes by integrating bulk and spatial sequencing data

Exploration between phenotypes and features

SpaLinker provides the *PhenoAssoFeatures* function to identify annotation or phenotype-related features in ST or other profiles. This function utilizes various methods that can be selected based on the type of the target variable. For clinical survival data, *PhenoAssoFeatures* employs the Cox regression and checks the proportional hazards assumption to ensure the validity of results. For binary group indicators, such as label for spots or samples, logistic regression, Wilcoxon Rank Sum test, DESeq2,⁸⁰ and AUC test are utilized. Continuous dependent variables, such as tumor proportion, TLS, and factor scores, are analyzed using correlation tests and linear regression. Except for DESeq2, the calculation is performed separately for each feature in the given data. The selection of significantly associated features is based on criteria such as *p* values, fold changes, or AUC powers ($(\text{AUC} - 0.5) \times 2$).

Determination of phenotype-related factors by NMF

NMF is an established machine learning method widely used for elucidating biologically relevant patterns or programs from transcriptomic data.^{44–46,108} In SpaLinker, NMF analysis is carried out to identify phenotype-related factors using bulk RNA-seq data combined with clinical information. As illustrated in Figure 5A, given an expression matrix with phenotype annotations, gene filtering is performed first using the *PhenoAssoFeatures* function to select phenotype-related genes. Then the filtered expression matrix is converted to a non-negative matrix by calculating the z-score for each feature and setting the negative values to zero. For prepared expression matrix E ($n \times m$), consisting of n genes and m samples, NMF analysis produced two matrices: the basis matrix W ($n \times k$), which encodes reference profiles of k factors, and the coefficients matrix H ($k \times m$), which represents the relative abundance of factors in the corresponding samples:

$$E = W \times H \quad (12)$$

The number of NMF factors (k) is set using the parameter rank, which can be determined by the users based on their prior knowledge or selected by SpaLinker. Specifically, if no rank is provided, SpaLinker uses an automatic method to select the appropriate value. First, NMF is run with different ranks ranging from 2 to 10 (by default) using the NMF (Version 0.28) R package. For each rank value, a default of 30 runs were performed, allowing the computation of a consensus matrix and cophenetic coefficient, which help visualizing and measuring the stability of the clusters obtained. Then, the coefficient reduction between every rank and next neighbor were calculated (the last one except). The rank with largest coefficient reduction and upper 0.95 (by default) was selected as the number of NMF factors. In SpaLinker, the NMF analysis and rank selection step can be completed with *RunNMFtest* function. Finally, phenotype-related factors are defined as clusters whose relative abundances in matrix H are significantly correlated with the phenotype, employing the same selection method used for genes.

Recovery of factors in spatial transcriptomic data

To establish a connection between the phenotype-related NMF factors and spatial characteristics of the TME, SpaLinker recovers these factors in ST data using the same methodology as described in EcoTyper.⁴⁵ First, for a new expression matrix E^* , it is converted into a non-negative matrix. Then the corresponding coefficients matrix H^* is determined using the previously fitted basis matrix W . This process is iteratively updated until convergence is achieved:

$$E^* = W \times H^* \quad (13)$$

In SpaLinker, this step is conducted using the *PredNMFInST* function.

Simulation of pseudo-bulk profiles and determination of TLS-related factors

To obtain the pseudo-bulk RNA-seq profiles with various TLS content, we used four RCC ST samples (frozen_b_1, frozen_b_18, frozen_a_3, and frozen_a_15) containing mature TLS. These samples were derived from frozen tumor tissues and sequenced in an unbiased manner to capture the full transcriptomic profiles. A total of 6429 non-TLS and 299 TLS spots were used. To remove batch effects across samples, we performed integrative analysis first using Seurat package (Figure S7A). Next, 50 TLS-negative pseudo-bulk profiles were generated by randomly sampling 1000 spots with replacement from non-TLS spots. Simultaneously, 450 TLS-positive pseudo-bulk profiles, with TLS content ranging from 1% to 90%, were generated by randomly selecting 1000 spots with replacement from both TLS and non-TLS spots. The expression levels were then averaged across spots to create new profiles. This process

resulted in 500 pseudo-bulk RNA-seq profiles with varying TLS content. TLS-related genes were identified by selecting genes with an absolute Pearson correlation coefficient greater than 0.5. NMF analysis was conducted on the filtered pseudo-bulk RNA-seq profiles, using TLS content as the phenotype and setting the rank to 2.

Determination of tumor-related and normal-related factors in LIHC cohort

In this study, the transcriptomic data and clinical data of LIHC and KIRC cohorts (TCGA) were downloaded from the USCS XENA portal (<https://xena.ucsc.edu/>).¹⁰⁹ In this study, only protein-coding, Immunoglobulin (Ig) and T-cell receptor (TCR) genes were retained for downstream analysis. Genes with expression greater than 1 in at least 20 percent samples were selected. To evaluate the robustness of NMF-derived factors, the expression profile was subjected to a resampling procedure. Subsets of the original dataset (424 samples) were randomly sampled at ratios ranging from 0.05 to 1.0. To ensure sufficient representation of normal samples, a constraint was imposed to retain at least 10 normal samples in subsets with a sampling ratio of 0.1 and 0.05. For each generated subset, differentially expressed genes (DEGs) between tumor and normal tissues were identified using the *PhenoAssoFeatures* function with ‘deseq2’ methods. Genes with *p* values less than 0.01 and a log₂ fold change greater than 1.5 were selected as tumor-related DEGs. NMF analysis was performed for each filtered expression profile, using rank parameters of 2, 3, and 4. For each identified tumor-related factor, we assessed its association with tumor tissue utilizing the predictive capability of the AUC metric. The AUC power is defined as $(\text{AUC} - 0.5) \times 2$.

Determination of TNI groups in RCC dataset

To investigate the conservation of TNI across 22 RCC samples, for each sample, the TNI regions were classified into various types based on their domain compositions and only TNI types that had more than 25 spots were considered. To enable comparable characterization across TNI types, the AUC power was used to assess the specific enrichment or depletion of genes in each TNI type. For each TNI type, ROC analysis was conducted to compare spots between the target TNI type and non-TNI regions within the same sample. Genes with an AUC power of 0.5 or higher were considered significantly enriched in TNI types. Subsequently, we identified 31 genes that were consistently enriched in at least 8 TNI types as TNI-specific genes, and only 54 TNI types that enriched at least one TNI-specific genes were used for analysis. This approach is also applied to immune and cancer related features. Using unsupervised hierarchical clustering based on the AUC powers of TNI-specific genes, we categorized these TNI types into four distinct groups. Unsupervised hierarchical clustering and visualization of clusters were performed using R package ComplexHeatmap⁷⁹ (Version 2.14.0).

Determination of survival-related factors in KIRC and LIHC cohorts

Survival-related genes from the KIRC and LIHC cohorts were identified using the *PhenoAssoFeatures* function with the ‘cox’ method. These genes had a significant correlation with survival ($p < 0.01$) and passed the proportional hazards assumption test ($p > 0.05$). In the Cox proportional-hazards regression models, a positive coefficient indicates a worse prognosis, while a negative coefficient indicates improved survival. Given the interference between genes with different effects, these genes were grouped as phenotype positive-related or negative-related types according to the direction of coefficients. Subsequently, NMF analysis was performed for the two gene groups respectively. The number of factors presented by rank parameter was selected according to the consensus matrix and cophenetic coefficient. Specifically, factors lacking a significant correlation with survival were excluded from subsequent analysis.

Determination of immune therapy response related factors in RNA-seq profiles of melanoma

To investigate the TME features associated with response to ICB therapy, we collected a cohort of melanoma patients who were treated with nivolumab (an anti-PD-1 agent).⁶³ A total of 98 samples with known labels were used, grouped into responders ($n = 20$; complete response [CR] or partial response [PR]) and non-responders ($n = 78$; progressive disease [PD] or stable disease [SD]). Therapy-related genes were identified using the *PhenoAssoFeatures* function with parameters set to method = “glm” and family = “binomial”. Genes with *p* values less than 0.05 were classified as response-positive or response-negative groups based on their coefficients. For each gene group, SpaLinker determined two significant NMF factors by setting the rank parameter to 2, resulting in a cophenetic coefficient above 0.995.

Functional enrichment analysis of top meta-genes for NMF factors

To clarify the functional pathways involved in the phenotype-related factors, we further analyzed each factor by selecting the top ranked genes based on their coefficients in the basis matrix *W*. The number of selected genes was set to range from 50 to 200, according to the number of phenotype-related genes. The pathway enrichment analysis was performed using the clusterProfiler (Version 4.6.2)⁸¹ R package, which provides enrichment analysis based on Reactome⁹⁸ pathway database. Pathways with *p* values less than 0.05 were considered significantly enriched.

Data visualization and survival analysis

ST visualization

SpaLinker enhances the visualization of spatial features by implementing various functions. Spatial feature plots, including cell abundances, gene expression, TLS scores, and domain labels, were obtained using the *SpotVisualize* function. Pie charts illustrating cell proportions in spots were produced with the *PlotCellSpot* function. Graphs displaying both tumor cell abundances and the defined TNI regions were generated via the *AbunBounPlot* function by inputting coordinate information, tumor cell abundances, and TNI labels. These functions were written based on the R package ggplot2⁸² (Version 3.4.0). Additionally, the creation of boxplots, bar plots,

density plots, and lollipop charts was accomplished through ggplot2 (Version 3.4.0). Moreover, the comparison of differential features was visualized with heatmaps generated using the ComplexHeatmap⁷⁹ R package (Version 2.14.0).

Survival analysis

To confirm the association between identified NMF factors and disease-specific survival, univariable Cox proportional-hazards regression models were performed using the relative proportions of each factor from the **H** matrix. Hazard ratios and *p* values were calculated using the *coxph* function in the survival R package (Version 3.5-7). Kaplan–meier survival plots for gene *TMSB4X* and TLS signatures were generated using the *ggsurvplot* function of the survminer R package (Version 0.4.9),⁸⁴ and log-rank test was used to calculate *p* values between groups.

QUANTIFICATION AND STATISTICAL ANALYSIS

The comparison of LR interaction scores between a single domain and the remaining spots were conducted using a two-sided Wilcoxon rank sum test. The *p* values were adjusted using the Benjamini-Hochberg (BH) method. Interactions with adjusted *p* values less than 0.01 were considered as significant expressed. The significance of the number of TLS-adjacent TNI types between TNI groups was assessed using Fisher's exact test. The significance levels of *p* values are as follows: ns (not significant), * <0.05, ** <0.01, *** <0.001, **** <0.0001. More detailed quantitative and statistical analyses are described in the relevant sections of the [STAR Methods](#) and in the figure legends. All statistical analyses were conducted using R.